

simple calculations reveal that, unlike the Pauli-X gate, the Pauli-Y and Pauli-Z gates not only affect the computational basis states but also introduce changes in their quantum phase.

$$Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \quad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

Figure 3: Matrix representations of the Pauli-Y and Pauli-Z gates

What happens at the hardware level when a Pauli-Z gate is applied to a superconducting qubit? Unlike a Pauli-X gate, which is implemented by applying a resonant microwave pulse that physically drives the qubit between the $|0\rangle$ and $|1\rangle$ states, a Pauli-Z gate does not move the qubit from one state to the other. The qubit remains in the same energy level, while only the relative phase between $|0\rangle$ and $|1\rangle$ is modified. In most modern superconducting quantum processors, this operation is implemented as a virtual Z gate. Rather than sending an additional microwave pulse to the qubit, the control electronics simply update the phase reference of all subsequent microwave pulses by the required angle. Since no physical pulse is applied, the operation is effectively instantaneous, introduces virtually no additional decoherence, and contributes negligible gate error. As a result, the Pauli-Z gate is one of the fastest and highest-fidelity quantum operations available on today's superconducting quantum computers.

Let us now turn our attention to the Pauli-Y gate, perhaps the most challenging of the three Pauli gates to understand. We have already seen its mathematical action through matrix multiplication, but what does the gate actually do? To build an intuitive understanding, let us temporarily sacrifice a little mathematical rigour. A useful way to think about the Pauli-Y gate is that it combines the effects of the Pauli-X and Pauli-Z gates. In other words, it performs both a bit flip and a phase flip simultaneously. Recall that the

Pauli-X gate flips the qubit from $|0\rangle$ to $|1\rangle$ (and vice versa) by performing a 180° rotation about the x-axis of the Bloch sphere. The Pauli-Z gate, on the other hand, leaves the computational basis states unchanged but introduces a 180° phase shift by rotating the qubit about the z-axis. The Pauli-Y gate combines these two ideas into a single operation, corresponding to a 180° rotation about the y-axis of the Bloch sphere.

Now, let us observe the effect of the Pauli-Y gate on the Bloch sphere. Figure 4 shows the evolution of the qubit when the Pauli-Y gate is applied to the state $|0\rangle$. Notice that the final state is again $|1\rangle$, just as it was for the Pauli-X gate discussed in the previous article. However, the two gates reach this state by following entirely different paths on the Bloch sphere. The Pauli-X gate performs a 180° rotation about the x-axis, whereas the Pauli-Y gate performs a 180° rotation about the y-axis.

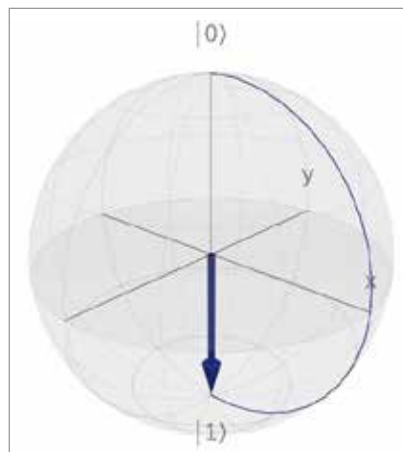


Figure 4: Effect of the Pauli-Y gate on $|0\rangle$

Does the path taken really matter? To build some intuition, let me use a simple analogy (assuming you are familiar with the geography of India). Suppose you and your friend are both travelling from Delhi to Chennai. Your flight goes via Kolkata, while your friend's flight goes via Mumbai. By evening, when someone asks where the two of you are, the answer is the same—you have both reached Chennai.

However, if someone asks at noon how far each of you is from Guwahati, the answers will be quite different. Although your destinations are identical, the routes you followed were not.

The Pauli-X and Pauli-Y gates behave in a similar manner. Starting from $|0\rangle$, both eventually reach $|1\rangle$, but the state vector follows a different trajectory on the Bloch sphere. For the basis states $|0\rangle$ and $|1\rangle$, this difference is not directly observable. However, for qubits in superposition states, the different rotations performed by the Pauli-X and Pauli-Y gates produce different relative phases, which can significantly influence the outcome of subsequent quantum operations. Understanding this subtle role of phase is one of the keys to understanding why quantum computing is fundamentally different from classical computing. As this is a challenging concept, we shall revisit it several times in the articles that follow in this series.

Now, let us examine the effect of the Pauli-Y gate on a superconducting qubit at the hardware level. Unlike the Pauli-X gate, which is implemented using a resonant microwave pulse that rotates the qubit by 180° about the x-axis, and the Pauli-Z gate, which is typically realized as a virtual Z gate by simply updating the phase reference of subsequent microwave pulses, the Pauli-Y gate requires both operations. In practice, it is implemented by applying a microwave pulse whose phase is shifted by 90° relative to that used for the Pauli-X gate, producing a 180° rotation about the y-axis of the Bloch sphere. Equivalently, the operation may be viewed as a suitable combination of Pauli-X and Pauli-Z operations. Consequently, the Pauli-Y gate performs both a bit flip and a phase flip simultaneously. Although its implementation is slightly more involved than that of the Pauli-X gate, it remains a high-fidelity operation on modern superconducting quantum processors.

Before concluding this article, let us briefly examine one of the most